

A PROJECT REPORT
ON
**UBER INDIA OPERATIONAL DATA
INTELLIGENCE**



Table of Contents

Chapter 1.....	2
1.1 Introduction	3
1.2 Need of the study.....	3
1.3 Objective	4
1.4 Research methodology.....	4
1.5 Limitations of the study	5
1.6 Future scope of the study	5
Chapter 2	6
2.1 Ride-Sharing Platforms and Platform Economics.....	7
2.2 Demand Forecasting and Supply Optimization in Transportation.....	7
2.3 Business Intelligence, Data Analytics, and Dashboard Design.....	8
2.4 Indian Ride-Sharing and Urban Mobility Market.....	8
Chapter 3.....	9
3.1 Platform Economics and the Gig Economy.....	10
3.2 Demand-Supply Balancing in Transportation Networks.....	10
3.3 Business Intelligence and Data-Driven Decision Making.....	11
3.4 Relational Database Theory and SQL.....	12
3.5 Data Visualization Theory.....	13
3.6 The Indian Ride-Sharing Market Context.....	14
Chapter 4	15
4.1 City-Level Demand and Revenue Analysis.....	16
4.2 Temporal Demand Analysis - Peak Hours and Time Slots.....	17
4.3 Surge Pricing Analysis.....	19
4.4 Driver Performance Analysis.....	20
4.5 Dataset Overview and Descriptive Statistics.....	23
4.6 Visualization.....	24
Chapter 5	33
5.1 Major Findings.....	34
5.2 Suggestions and Recommendations.....	35
5.3 Conclusion.....	38
References.....	39

Chapter – 1

INTRODUCTION

1.1 Introduction

The rapid growth of digital platforms has transformed the way people move across cities. Ride-sharing applications such as Uber have fundamentally altered urban transportation by connecting riders and drivers through a technology-driven marketplace, offering convenience, affordability, and real-time availability at the tap of a button. India, with its vast urban population of over 500 million city dwellers and a growing smartphone user base, has emerged as one of the most important and complex markets for app-based ride-sharing services in the world. Uber entered the Indian market in 2013 and has since expanded its presence across more than 100 cities, offering a diverse portfolio of vehicle categories including UberGo, UberX, UberXL, UberAuto, and UberMoto the latter two being uniquely Indian additions that reflect the country's distinct transportation culture. When demand exceeds supply, riders face long wait times, surge pricing, or no car availability leading to cancellations and customer dissatisfaction. When supply exceeds demand, drivers sit idle and earn less leading to driver attrition and reduced platform loyalty. Both scenarios erode the platform's value on both sides of the marketplace. This project addresses this visibility gap by building an end-to-end business analytics solution. Using a dataset of 300 Uber India trip records across 10 cities and 50 drivers, the project employs MySQL for structured data storage, SQL for analytical querying, and Microsoft Power BI for interactive dashboard visualization. The result is an operational dashboard that enables Uber India's management teams to monitor city performance, identify peak demand windows, detect surge pricing patterns, and evaluate driver quality all from a single, connected analytical platform.

1.2 Need for the Study

The need for this study arises from a clear and measurable gap in Uber India's operational performance. Preliminary analysis of the dataset used in this project reveals that 27.7% of all trip requests 83 out of 300 go unserved. Of these, 46 trips are cancelled and 37 result in 'No Car Available,' meaning that more than one-quarter of all demand on the platform fails to convert into a completed ride. This represents a significant loss of revenue, customer goodwill, and competitive positioning. Despite the availability of large volumes of operational data, many ride-sharing platforms including Uber at the city operations level continue to rely on fragmented, delayed, or anecdotal information for day-to-day decision-making. City managers may not have immediate visibility into which zones are undersupplied, which hours are generating unmet demand, or which drivers are consistently underperforming. Without a structured analytical framework and an interactive dashboard, these insights remain buried in raw data and inaccessible to the decision-makers who need them most. Furthermore, existing academic and industry research on ride-sharing analytics is largely focused on US and European markets. The Indian ride-sharing market with its unique night-economy demand patterns, diverse vehicle categories, and multi-tier city structure presents distinct analytical challenges that generic global studies do not address. This project fills that gap by applying structured business analytics specifically to Indian city and driver data, producing findings and recommendations that are directly relevant to Uber's India operations.

1.3 Objectives of the Study

Main Objective

To design and implement an end-to-end business analytics solution comprising a MySQL relational database, SQL analytical queries, and a Power BI interactive dashboard that identifies demand-supply imbalances in Uber India's ride-sharing operations across 10 cities and provides actionable recommendations for operational optimization.

Sub-Objectives

1. **Database Design:** To design and implement a normalized relational database in MySQL containing trip records, driver profiles, and city information for 10 Indian cities, structured as a star schema to support efficient analytical querying.
2. **SQL Analysis:** To write 20 structured SQL queries across five analytical modules dataset overview, city performance, temporal demand patterns, surge pricing analysis, and driver performance to extract quantified business insights from the raw data.
3. **Demand Pattern Identification:** To identify peak demand hours, busiest time slots, and geographic zones with highest trip volumes to support time-based and location-based driver scheduling decisions.
4. **Surge Pricing Analysis:** To examine surge pricing activation patterns across cities and pickup zones, quantify the revenue premium generated by surge events, and identify the locations and times where surge is most frequently triggered.
5. **Driver Performance Evaluation:** To evaluate driver quality using rating and experience metrics, classify drivers into performance categories, and identify underperforming markets requiring supply-side intervention.
6. **Power BI Dashboard:** To build a five-page interactive Power BI dashboard that translates SQL findings into visual form, enabling non-technical operations teams to monitor KPIs, explore city comparisons, and identify demand-supply gaps at a glance.

1.4 Research Methodology

The dataset used in this project is the Uber India Dataset — a structured sample dataset containing 300 trip records across 10 Indian cities (Ahmedabad, Delhi, Chennai, Kochi, Mumbai, Hyderabad, Bengaluru, Kolkata, Jaipur, and Pune), 50 driver profiles, and 10 city records stored in Microsoft Excel format (.xlsx) and loaded into a MySQL 8.0 relational database for analysis.

The analytical tools and techniques used are as follows:

- **Data Storage:** MySQL 8.0 — relational database with star schema design comprising three tables: trips (fact table), drivers, and cities (dimension tables).
- **Data Analysis:** Structured Query Language (SQL) — 20 queries using SELECT, WHERE, GROUP BY, ORDER BY, JOIN, CASE WHEN, COUNT, SUM, AVG, ROUND, and LIMIT clauses across five analytical modules.
- **Data Visualization:** Microsoft Power BI Desktop — five-page interactive dashboard with DAX measures, KPI cards, bar charts, donut charts, matrix heatmaps, and slicers.

- **Data Preparation:** Microsoft Excel initial dataset exploration, validation of column data types, and CSV export for MySQL import.

1.5 Limitations of the Study

Every research project operates within certain boundaries, and this study is no exception. The following limitations should be considered when interpreting the findings and recommendations presented in this report.

- **Static Data:** The analysis is based on historical data from January to March 2024 only. The Power BI dashboard provides historical views and does not reflect real-time operational conditions, as live Uber API data is not publicly accessible.
- **Geographic Coverage:** Only 10 Indian cities are included in the dataset. The findings may not generalize to all 100+ cities where Uber India operates, particularly smaller Tier-2 and Tier-3 markets with distinctly different demand profiles.
- **Driver Behavioural Data:** The dataset does not include driver-side metrics such as online hours, idle time, GPS location, or response time. These would significantly enrich the supply-side analysis and enable more precise driver repositioning recommendations.
- **External Factors:** The study does not control for external influences such as weather conditions, public holidays, local events, or competitor promotions that may have affected trip demand and cancellation patterns during the study period.

1.6 Future Scope of the Study

This project establishes a strong analytical foundation that can be extended in several valuable directions. Future research could integrate machine learning models such as time-series forecasting or gradient boosting to predict hourly demand by city zone, enabling proactive driver positioning rather than reactive surge pricing. The current static Power BI dashboard could be evolved into a real-time operational monitoring system by connecting it to a live data pipeline refreshed every 5-10 minutes using tools such as Apache Kafka or Azure Stream Analytics. Expanding the dataset to include all 100+ Uber India cities, along with customer demographic data and competitor trip records, would enable a comprehensive market share and customer segmentation analysis across India's diverse urban landscape. Finally, the analytical framework developed in this project is directly transferable to other two-sided marketplace platforms such as food delivery, logistics, and healthcare appointment services, making it a broadly applicable template for demand-supply optimization analytics.

Chapter 2

Literature Review

2.1 Ride-Sharing Platforms and Platform Economics

Rochet and Tirole (2003) established the foundational theory of two-sided markets in their seminal paper 'Platform Competition in Two-Sided Markets,' published in the Journal of the European Economic Association. They argued that platforms serving two distinct user groups such as buyers and sellers, or in the case of Uber, riders and drivers must price and manage

both sides simultaneously. **Parker, Van Alstyne, and Choudary (2016)** extended this framework in their widely cited book 'Platform Revolution,' arguing that platform businesses must actively orchestrate interactions between participants rather than passively facilitating them. Their concept of the 'demand-supply feedback loop' where more drivers attract more riders and vice versa is central to this project's premise that supply-side gaps in specific cities and time windows break the platform's core value proposition. **Hall and Krueger (2018)** conducted a landmark empirical study of Uber's driver labor market titled 'An Analysis of the Labor Market for Uber's Driver-Partners in the United States,' published in the ILR Review. Their research found that driver supply is highly elastic with respect to earnings drivers come online when demand is high and surge pricing is active, and go offline when expected earnings fall.

2.2 Demand Forecasting and Supply Optimization in Transportation

Bimpikis, Candogan, and Saban (2019) in their paper 'Spatial Pricing in Ride-Sharing Networks,' published in Operations Research, developed a mathematical model demonstrating that urban geography significantly determines optimal pricing and supply strategies for ride-sharing platforms. Their key finding that zones with concentrated demand generators such as airports and commercial districts require fundamentally different strategies than residential zones is directly validated by this project's analysis. **Ke, Zheng, Yang, and Chen (2017)** proposed a spatio-temporal deep learning model for short-term passenger demand prediction in their paper 'Short-Term Forecasting of Passenger Demand Under On-Demand Ride Services,' published in Transportation Research Part C. Their model incorporated both spatial dependencies between zones and temporal patterns across hours and days of the week, achieving significantly better prediction accuracy than ARIMA-based baselines. **Castillo, Knoepfle, and Weyl (2017)** studied what they termed the 'Wild Goose Chase' problem in ride-sharing, presented at the ACM Conference on Economics and Computation. They found that when surge pricing incentivizes too many drivers to reposition toward high-demand zones simultaneously, supply can paradoxically overshoot demand creating inefficiency at the platform level even while appearing to solve a local imbalance. **Talluri and van Ryzin (2004)** in their foundational textbook 'The Theory and Practice of Revenue Management,' formalized dynamic pricing as a revenue management optimization problem. They established that platforms should set prices at each moment to maximize total expected revenue across the entire demand horizon.

2.3 Business Intelligence, Data Analytics, and Dashboard Design

Negash (2004) defined Business Intelligence in the Communications of the Association for Information Systems as systems that 'combine data gathering, data storage, and knowledge management with analytical tools to present complex internal and competitive information to planners and decision-makers.' His four-layer BI framework data sources, data storage, analytical processing, and presentation maps precisely onto the architecture of this project: Excel data sources, MySQL storage, SQL analytical processing, and Power BI presentation.

Few (2006) in his influential work 'Information Dashboard Design,' established that effective operational dashboards must present the most important information needed to achieve one or more objectives, arranged on a single screen so it can be monitored at a glance. His principles of perceptual efficiency eliminating visual clutter, using color only to encode meaning, and organizing information by decision priority directly informed the design of this project's five-page Power BI dashboard. **Kimball and Ross (2013)** in 'The Data Warehouse Toolkit,' established the star schema as the preferred dimensional modeling approach for analytical databases intended to feed business intelligence tools. Their argument that separating measurable facts from descriptive dimensions simplifies both query writing and BI tool integration is the theoretical basis for the database design used in this project. The trips table serves as the fact table containing measurable events, while the drivers and cities tables serve as dimension tables providing descriptive context.

2.4 Indian Ride-Sharing and Urban Mobility Market

Agarwal, Mani, and Telang (2020) studied the impact of Uber's entry on taxi markets in Indian cities, finding that app-based ride-sharing significantly improved service quality and price transparency compared to traditional auto-rickshaws and radio taxis, particularly for airport and long-distance urban trips. Their finding that airport trips represent a disproportionately high share of ride-sharing revenue in Indian cities aligns with this project's empirical result that airport pickups account for 29% of total revenue despite only 12.9% of trip volume confirming that the airport revenue premium is a structural feature of Indian ride-sharing markets, not a dataset-specific anomaly. **Sood and Nagpal (2019)** analyzed the factors driving ride-sharing adoption in Indian Tier-1 cities in their study published in the Journal of Urban Economics. They found that late-night availability, surge pricing transparency, and driver quality ratings were the three most influential factors determining customer loyalty. **Ministry of Road Transport and Highways, Government of India (2023)** in its annual report on urban transportation noted that India's app-based taxi and ride-sharing sector grew by 34% between 2021 and 2023, with metro cities accounting for 68% of total trips and Tier-1 cities like Ahmedabad, Pune, and Jaipur showing the fastest growth rates. This policy context directly supports the geographic scope of this project's dataset which includes both established metro markets and growing Tier-1 cities and validates the strategic importance of understanding demand patterns in both market tiers simultaneously.

Chapter 3

Theoretical Framework

3.1 Platform Economics and the Gig Economy

Two-Sided Market Theory

Uber operates as a classic two-sided market a platform that serves two distinct user groups simultaneously: riders who demand transportation services and drivers who supply them. The concept of two-sided markets was formally established by Rochet and Tirole (2003) and later expanded by Parker and Van Alstyne (2005), who argued that the value of a platform increases for each side as the number of participants on the other side grows. This is commonly referred to as the network effect. In Uber's context, more drivers on the platform reduce waiting times for riders, which attracts more riders. More riders generate more trip opportunities for drivers, which attracts more drivers. This self-reinforcing cycle known as a positive feedback loop is the fundamental growth engine of any ride-sharing platform. However, the same dynamic that powers growth also creates operational vulnerability: any imbalance between supply (drivers) and demand (riders) breaks the cycle and erodes platform value on both sides simultaneously.

The Gig Economy and Labor Supply Elasticity

Uber drivers operate as independent contractors within the gig economy a labor market characterized by flexible, short-term engagements rather than permanent employment contracts. Hall and Krueger (2018) studied Uber's driver labor market and found that driver supply is highly elastic with respect to earnings opportunities: drivers go online when expected earnings are high and offline when expected earnings are low. This elasticity is both a strength and a challenge for platform operations. The strength is that surge pricing Uber's mechanism for raising fares during high demand automatically incentivizes drivers to come online, partially self-correcting supply shortages. The challenge is that driver behavior is unpredictable in the short term: drivers may not respond to surge signals quickly enough to prevent service deterioration, particularly in markets like Kolkata and Chennai where, as our dataset reveals, cancellation rates reach 23.1% and 17.6% respectively.

3.2 Demand-Supply Balancing in Transportation Networks

Spatial Demand Distribution

Transportation demand is inherently spatial, it varies by location as much as by time. Bimpikis, Candogan, and Saban (2019) demonstrated through mathematical modeling that urban geography significantly determines the optimal pricing and supply strategy for ride-sharing platforms. Their key finding was that city zones with concentrated demand generators such as airports, commercial districts, and transit hubs require fundamentally different operational strategies than residential zones with dispersed, unpredictable demand. This spatial dimension is central to our project's analytical design. The trips dataset contains a pickup_area column that captures not just the city but the specific neighborhood or zone of each trip request. Our analysis reveals a stark spatial concentration: airport pickup zones in Mumbai, Delhi, Chennai, Ahmedabad, and other cities generate average fares of ₹93.08 per trip 2.75 times higher than the ₹33.84 average for regular pickup zones. This extreme spatial concentration of premium demand is precisely the pattern predicted by spatial demand distribution theory.

Temporal Demand Patterns and Peak Load Theory

Transportation demand exhibits strong temporal cyclicalities, it follows predictable patterns across hours of the day, days of the week, and seasons of the year. This temporal structure is the basis of peak load pricing theory, originally developed by Boiteux (1960) for utility industries and later applied to transportation by Vickrey (1969). The core insight of peak load theory is that prices should be higher during peak demand periods both to manage demand and to fund the additional capacity needed to serve peaks. Uber's surge pricing is a direct application of peak load pricing theory fares rise during periods of high demand to both ration scarce driver supply and incentivize additional supply to enter the market. Our analysis of the Uber India dataset reveals that the peak demand periods in Indian cities do not fully conform to the traditional morning and evening rush hour pattern. Instead, our data shows that Late Night and Early Morning (midnight to 5 AM) is the single busiest time window with 58 completed trips, suggesting that Indian urban ride-sharing exhibits a distinct night-economy peak pattern driven by entertainment, dining, and late-shift work activities.

Dynamic Pricing Theory

Dynamic pricing, adjusting prices in real time based on current demand and supply conditions is a well-established theoretical framework in economics and operations research. Talluri and van Ryzin (2004) formalized dynamic pricing as a revenue management optimization problem: the platform sets prices at each moment to maximize total expected revenue across the entire demand horizon, not just the current moment. Uber's surge pricing algorithm is an implementation of dynamic pricing theory. When the ratio of ride requests to available drivers in a given zone exceeds a threshold, the surge multiplier is applied raising fares by 1.5x, 2.0x, 2.5x, or 3.0x. Our dataset shows that 82 out of 217 completed trips (37.8%) were surged, with airport zones producing the highest surge multipliers consistently. The theoretical implication is that surge pricing is functioning as designed in airport zones clearing excess demand but may not be sufficient in residential areas where 12.3% of all requests result in 'No Car Available' outcomes despite no surge being applied.

3.3 Business Intelligence and Data-Driven Decision Making

The Business Intelligence Framework

Business Intelligence (BI) refers to the set of technologies, processes, and methodologies that organizations use to collect, integrate, analyze, and present business data for the purpose of supporting better decision-making. Negash (2004) defined BI as systems that combine data gathering, data storage, and knowledge management with analytical tools to present complex internal and competitive information to decision-makers. The modern BI stack typically consists of four layers: data sources, data storage, analytical processing, and visualization. This project implements a complete four-layer BI stack specifically for Uber's India operations. The data source layer consists of three structured tables — trips, drivers, and cities — derived from the Uber India dataset. The data storage layer is a normalized relational database in MySQL with a star schema design. The analytical processing layer consists of 20 SQL queries that transform raw data into business metrics.

The CRISP-DM Analytical Methodology

The Cross-Industry Standard Process for Data Mining (CRISP-DM) is the most widely adopted methodology for structuring data analytics projects. Developed by a consortium of European companies in 1999 and later formalized by Wirth and Hipp (2000), CRISP-DM defines a six-phase cyclical process: Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment. This project follows the CRISP-DM framework explicitly across its phases. In the Business Understanding phase, we identified the core operational problem demand-supply imbalance across Indian cities and defined five measurable project objectives. In the Data Understanding phase, we explored the trips, drivers, and cities datasets to understand their structure, quality, and analytical potential. In the Data Preparation phase, we cleaned and loaded the data into MySQL with appropriate data types and indexes. In the Modeling phase, we wrote 20 SQL queries implementing four analytical modules. In the Evaluation phase, we validated query outputs against expected business logic. In the Deployment phase, we connected the database to Power BI and built the interactive dashboard.

Key Performance Indicators (KPI) Theory

A Key Performance Indicator is a measurable value that demonstrates how effectively an organization is achieving its key business objectives. Parmenter (2015) distinguished between Result Indicators which tell you what has happened and Performance Indicators which tell you what to do to improve results. Effective KPI design requires that indicators be Specific, Measurable, Achievable, Relevant, and Time-bound the SMART framework.

This project defines and calculates eight core KPIs for Uber India operations, each designed to be directly actionable by operations teams. The Completion Rate (72.3%) tells management how much demand is being successfully served. The Cancellation Rate (15.3%) identifies service reliability issues by city. The Average Surge Multiplier (1.19x) measures pricing efficiency. The Revenue per City ranking identifies investment priorities.

3.4 Relational Database Theory and SQL

The Relational Model

The relational model of data management was introduced by Edgar F. Codd in his landmark 1970 paper 'A Relational Model of Data for Large Shared Data Banks.' Codd proposed organizing data into relations (tables) consisting of tuples (rows) and attributes (columns), with relationships between tables expressed through shared key values rather than physical pointers. This model provides three fundamental guarantees that make it ideal for analytical projects: data integrity through constraints, query flexibility through SQL, and data independence from physical storage details. This project implements the relational model using MySQL 8.0. The three tables trips, drivers, and cities are connected through foreign key relationships: the driver_id column in the trips table references the driver_id primary key in the drivers table, and the city column in the trips table references the city_name in the cities table. These relationships enforce referential integrity ensuring that no trip can reference a driver or city that does not

exist in the system and enable the JOIN operations that are central to multi-dimensional analysis.

Star Schema Design for Analytical Databases

In this project, the trips table functions as the fact table it contains measurable events (each trip) with quantitative metrics (fare, surge, distance_km) and foreign keys pointing to dimension tables. The drivers table and cities table function as dimension tables they contain descriptive attributes (driver_name, rating, city_name, state, zone_type) that provide context for the facts. This dimensional structure is why our SQL queries can answer rich questions like 'What is the average fare by vehicle type across Metro cities?' by joining the fact table to multiple dimensions in a single query.

SQL as the Language of Data Analysis

Structured Query Language (SQL) was developed at IBM in the early 1970s based on Codd's relational algebra and has since become the universal standard for querying relational databases (ISO/IEC 9075). SQL's power as an analytical tool lies in its declarative nature: the analyst specifies what data is needed rather than how to retrieve it, and the database engine determines the optimal execution plan. This makes SQL accessible to business analysts without requiring knowledge of data structures or algorithms.

The 20 SQL queries written in this project span the full spectrum of SQL analytical capabilities. Basic queries use SELECT, FROM, WHERE, GROUP BY, and ORDER BY to extract and aggregate data. Intermediate queries use JOIN to combine data from multiple tables and CASE WHEN to create categorical variables. Advanced queries use aggregate functions (COUNT, SUM, AVG, MIN, MAX) with conditional logic to calculate business metrics like cancellation rates and surge distributions. Each query is designed to answer a specific business question embodying the principle that analytical SQL should be question-driven, not data-driven.

3.5 Data Visualization Theory

The Principles of Effective Data Visualization

Data visualization is the graphical representation of information and data. Tufte's core principle is maximizing the data-ink ratio the proportion of a graphic's ink devoted to non-redundant data. A good visualization shows the most data with the least unnecessary visual elements. Stephen Few (2006) extended these principles specifically to business dashboards, arguing that effective dashboards must present the most important information needed to achieve one or more objectives, consolidated and arranged on a single screen so the information can be monitored at a glance. Few's principles directly inform the design of this project's Power BI dashboard: each of the five pages is organized around a single business question, with visuals chosen for their ability to communicate specific data patterns bar charts for comparisons, line charts for trends, donut charts for proportions, and matrix heatmaps for two-dimensional patterns.

DAX and In-Memory Analytics

Data Analysis Expressions (DAX) is a formula language developed by Microsoft for Power BI, Power Pivot, and Analysis Services. DAX is built on the concept of the in-memory columnar database storing data by column rather than by row which enables extremely fast aggregations across large datasets. The five DAX measures created in this project, Total Trips, Completed Trips, Total Revenue, Avg Fare, and Avg Surge all use the CALCULATE function, which is DAX's mechanism for modifying the filter context. For example, the Completed Trips measure uses CALCULATE(COUNTROWS(trips), trips[trip_status] = 'Completed') to override the default filter context and count only rows where trip_status equals 'Completed,' regardless of what other filters are active on the report page. This context-awareness makes DAX measures far more powerful than simple column aggregations for interactive dashboard development.

3.6 The Indian Ride-Sharing Market Context

Structure of the Indian Urban Mobility Market

India's urban mobility market is one of the fastest-growing and most complex in the world. With over 500 million urban residents across cities ranging from tier-1 metros like Mumbai and Delhi to rapidly growing tier-2 cities like Jaipur and Kochi, India presents a unique operational environment for ride-sharing platforms. Unlike Western markets where personal car ownership is high, Indian urban commuters rely heavily on a multi-modal mix of public transport, auto-rickshaws, two-wheelers, and app-based ride-sharing services. Uber entered the Indian market in 2013 and has since expanded to over 100 Indian cities. Its India-specific vehicle categories UberAuto (auto-rickshaw) and UberMoto (motorcycle) reflect adaptations to local transportation culture that are not present in Uber's Western markets. The dataset used in this project includes all five vehicle types: UberGo, UberX, UberXL, UberAuto, and UberMoto providing a comprehensive view of Uber's full India product portfolio.

Data Analytics in Indian Transportation

The application of data analytics to transportation optimization in India is still an emerging practice. While global ride-sharing companies like Uber and Ola have sophisticated internal analytics platforms, academic and industry research specifically examining the analytical patterns of Indian ride-sharing operations remains limited compared to the extensive literature on US and European markets.

The 10 cities included in this project Mumbai, Delhi, Bengaluru, Hyderabad, Chennai, Pune, Kolkata, Ahmedabad, Jaipur, and Kochi represent a deliberately diverse sample spanning four Indian states in South India, three in West India, two in North India, and one in East India. This geographic diversity ensures that the analytical findings reflect genuinely different market conditions rather than the characteristics of a single region, making the project's conclusions applicable across Uber's real India operations.

Chapter 4

Data Analysis and Interpretation

This chapter presents the complete data analysis conducted on the Uber India dataset comprising 300 trip records, 50 drivers, and 10 cities across the January to March 2024 period. The analysis is organized into five analytical modules, each examining a distinct dimension of Uber's demand and supply performance. For each module, the SQL query used to extract the data is presented, followed by the results in tabular form, and a detailed interpretation of the findings. The Power BI dashboard visualizations corresponding to each analysis are referenced throughout.

The dataset covers ten Indian cities Ahmedabad, Delhi, Chennai, Kochi, Mumbai, Hyderabad, Bengaluru, Kolkata, Jaipur, and Pune spanning Metro, Tier-1, and Tier-2 market classifications. The five vehicle types covered are UberGo, UberX, UberXL, UberAuto, and UberMoto, reflecting Uber's full India product portfolio including India-specific categories.

4.1 City-Level Demand and Revenue Analysis

Trip Volume and Revenue by City

SQL Query

```
SELECT
  city,
  COUNT(*) AS total_requests,
  COUNT(CASE WHEN trip_status = 'Completed' THEN 1 END) AS completed,
  COUNT(CASE WHEN trip_status = 'Cancelled' THEN 1 END) AS cancelled,
  COUNT(CASE WHEN trip_status = 'No Car Available' THEN 1 END) AS no_car,
  ROUND(SUM(fare), 2) AS total_revenue,
  ROUND(AVG(fare), 2) AS avg_fare,
  ROUND(AVG(surge), 2) AS avg_surge
FROM trips
GROUP BY city
ORDER BY total_revenue DESC;
```

Results

City	Requests	Completed	Cancelled	No Car	Revenue (₹)	Avg Fare (₹)	Avg Surge
Ahmedabad	42	30	8	4	1,379.80	32.85	1.20
Delhi	38	30	6	2	1,358.29	35.74	1.24
Chennai	34	24	6	4	1,119.53	32.93	1.26
Kochi	30	22	4	4	894.20	29.81	1.22
Mumbai	28	20	4	4	841.24	30.04	1.16
Hyderabad	30	23	4	3	839.83	27.99	1.13
Bengaluru	32	23	4	5	760.29	23.76	1.17
Kolkata	26	17	6	3	689.08	26.50	1.15
Jaipur	19	14	1	4	637.80	33.57	1.21
Pune	21	14	3	4	482.32	22.97	1.10

Interpretation

Ahmedabad emerges as Uber India's highest revenue-generating city at ₹1,379.80, despite not being traditionally considered a primary ride-sharing market. This finding is driven by

Ahmedabad's combination of high trip volume (42 requests the highest among all cities) and consistent surge activation, reflecting strong underlying demand in Gujarat's commercial capital. Delhi follows closely at ₹1,358.29 with an identical completion count of 30 trips, and benefits from a slightly higher average fare of ₹35.74 compared to Ahmedabad's ₹32.85, suggesting longer average trip distances in the capital. Chennai presents a notable performance paradox it ranks third in revenue at ₹1,119.53 but has the second highest cancellation rate at 17.6% and four instances of 'No Car Available.' This suggests that Chennai has strong demand that the current driver supply is struggling to meet, making it a priority city for driver recruitment. Pune ranks last in revenue at ₹482.32 with the lowest average fare (₹22.97) and lowest surge multiplier (1.10x), indicating a market where Uber has not yet achieved sufficient demand density for efficient operation.

Key Findings – The top 3 cities Ahmedabad, Delhi, and Chennai contribute ₹3,857.62 (42.9%) of total platform revenue despite accounting for only 114 requests (38%) of total demand. This revenue concentration justifies prioritizing these markets for supply investment.

4.2 Temporal Demand Analysis - Peak Hours and Time Slots

Top 8 Busiest Hours of the Day

SQL Query

```
SELECT
  trip_hour,
  COUNT(*) AS total_trips
FROM trips
GROUP BY trip_hour
ORDER BY total_trips DESC
LIMIT 10;
```

Results - Top 8 Hours

Rank	Hour	Time (12hr)	Total Trips
1	21:00	9:00 PM	20
2	00:00	12:00 AM (Midnight)	19
3	12:00	12:00 PM (Noon)	18
4	13:00	1:00 PM	17
5	03:00	3:00 AM	16
6	09:00	9:00 AM	15
7	17:00	5:00 PM	15
8	04:00	4:00 AM	14

Trips and Fare by Time Slot

SQL Query

```
SELECT
  CASE
    WHEN trip_hour BETWEEN 6 AND 9 THEN 'Morning Rush (6-9 AM) '
    WHEN trip_hour BETWEEN 10 AND 16 THEN 'Afternoon (10AM-4PM) '
    WHEN trip_hour BETWEEN 17 AND 20 THEN 'Evening Rush (5-8 PM) '
    WHEN trip_hour BETWEEN 21 AND 23 THEN 'Night (9-11 PM) '
    ELSE 'Late Night / Early Morning'
  END AS time_slot,
  COUNT(*) AS trip_count,
  ROUND(AVG(fare), 2) AS avg_fare
FROM trips
WHERE trip_status = 'Completed'
GROUP BY time_slot
ORDER BY trip_count DESC;
```

Results

Time Slot	Completed Trips	% of Total	Avg Fare (₹)
Late Night / Early Morning (0–5 AM)	58	26.7%	41.19
Afternoon (10 AM – 4 PM)	52	24.0%	43.08
Evening Rush (5–8 PM)	42	19.4%	45.55
Morning Rush (6–9 AM)	33	15.2%	45.46
Night (9–11 PM)	32	14.7%	29.99

Interpretation

The temporal demand analysis produces one of the most significant and counterintuitive findings of this project. The single busiest hour across the entire dataset is 9 PM with 20 trips, followed by Midnight (12 AM) with 19 trips. Furthermore, the Late Night and Early Morning window (midnight to 5 AM) accounts for 58 completed trips the highest of any time slot at 26.7% of total completed trips. This finding strongly contradicts the conventional assumption that Indian ride-sharing demand peaks during morning and evening rush hours. The rush hour slots, while recording the highest average fares (Morning Rush at ₹45.46 and Evening Rush at ₹45.55), contribute only 15.2% and 19.4% of trips respectively significantly less than the Late Night window. The Night slot (9–11 PM) shows the lowest average fare at ₹29.99, suggesting these are shorter recreational trips to restaurants, markets, or social gatherings rather than the longer airport or office commute trips that characterize higher-fare windows.

Key Findings – Late Night and Early Morning (0–5 AM) is Uber India's busiest time window with 58 trips (26.7% of demand). This night-economy pattern is driven by India's vibrant late-night urban culture in cities like Mumbai, Delhi, Bengaluru, and Hyderabad.

4.3 Surge Pricing Analysis

Surge Level Distribution

SQL Query

```
SELECT
  CASE
    WHEN surge = 1.0 THEN 'No Surge'
    WHEN surge = 1.5 THEN 'Low Surge (1.5x)'
    WHEN surge = 2.0 THEN 'Medium Surge (2.0x)'
    ELSE 'High Surge (2.5x+)'
  END AS surge_level,
  COUNT(*) AS trip_count
FROM trips
GROUP BY surge_level
ORDER BY trip_count DESC;
```

Results

Surge Level	Trip Count	% of All Trips	Multiplier Range
No Surge	218	72.7%	1.0x
Low Surge (1.5x)	63	21.0%	1.5x
Medium Surge (2.0x)	12	4.0%	2.0x
High Surge (2.5x+)	7	2.3%	2.5x – 3.0x

Airport vs Regular Pickup Revenue Comparison

SQL Query

```
SELECT
  CASE
    WHEN pickup_area LIKE '%Airport%' THEN 'Airport Pickup'
    ELSE 'Regular Pickup'
  END AS pickup_type,
  COUNT(*) AS total_trips,
  ROUND(AVG(fare), 2) AS avg_fare,
  ROUND(SUM(fare), 2) AS total_revenue
FROM trips
WHERE trip_status = 'Completed'
GROUP BY pickup_type;
```

Results

Pickup Type	Completed Trips	% of Volume	Avg Fare (₹)	Total Revenue (₹)	Revenue Share
Airport Pickup	28	12.9%	93.08	2,606.20	28.9%
Regular Pickup	189	87.1%	33.84	6,396.18	71.1%

Top 5 Highest Fare Trips

SQL Query

```
SELECT trip_id, city, pickup_area, trip_date,
  fare, surge, distance_km
FROM trips
WHERE trip_status = 'Completed'
ORDER BY fare DESC
LIMIT 5;
```

Results

City	Pickup Area	Date	Fare (₹)	Surge	Distance (km)
Chennai	Airport (MAA)	2024-01-14	143.20	2.5x	25.4
Ahmedabad	Airport (AMD)	2024-02-04	129.11	2.0x	32.0
Delhi	Airport (DEL)	2024-01-19	126.19	3.0x	19.3
Jaipur	Airport (JAI)	2024-02-28	122.61	2.0x	32.0
Ahmedabad	Airport (AMD)	2024-02-17	111.24	3.0x	19.7

Interpretation

The surge analysis reveals that 27.3% of all trips in the dataset experienced some form of surge pricing. Of the 217 completed trips, 82 (37.8%) had a surge multiplier above 1.0x. This high surge frequency indicates that demand consistently outpaces supply in specific zones and time windows across Indian cities, triggering Uber's dynamic pricing algorithm.

The airport pickup analysis produces the most commercially significant finding in the entire project. Airport pickups account for only 28 out of 217 completed trips (12.9% of volume) yet generate ₹2,606.20 in revenue (28.9% of total). Each airport trip earns an average of ₹93.08 exactly 2.75 times the ₹33.84 average for regular pickups. This extraordinary revenue premium is driven by the combination of longer trip distances and higher surge multipliers that are consistently active at airport zones.

All five of the highest individual fares in the dataset originate from airport pickup zones, confirming that airports are Uber India's single most valuable location type. The maximum fare of ₹143.20 was recorded at Chennai Airport (MAA) on 14 January 2024, on a 25.4 km trip with a 2.5x surge multiplier. Delhi Airport's ₹126.19 fare on a 19.3 km trip at 3.0x surge demonstrates that surge multiplier is sometimes more impactful than distance in determining the final fare.

Key Findings – Airport pickups are worth 2.75x more per trip than regular pickups. Every 1 airport trip equals 2.75 regular trips in revenue terms. Airport zone supply management is the single highest-ROI operational lever available to Uber India.

4.4 Driver Performance Analysis

Driver Performance Categories

SQL Query

```
SELECT
  driver_name,
  city,
  vehicle_type,
  rating,
  total_trips,
  CASE
    WHEN rating >= 4.5 AND total_trips >= 1000 THEN 'Star Driver'
    WHEN rating >= 4.0 AND total_trips >= 500 THEN 'Good Driver'
    ELSE 'Needs Improvement'
  END AS driver_category
FROM drivers
WHERE status = 'Active'
ORDER BY rating DESC, total_trips DESC;
```

Results - Driver Category Summary

Category	Count	% of Active Drivers	Criteria
Star Driver	11	28.9%	Rating ≥ 4.5 AND Total Trips ≥ 1,000
Good Driver	15	39.5%	Rating ≥ 4.0 AND Total Trips ≥ 500
Needs Improvement	12	31.6%	Below Good Driver threshold

Revenue by Vehicle Type

SQL Query

```
SELECT
  d.vehicle_type,
  COUNT(t.trip_id) AS total_trips,
  ROUND(SUM(t.fare), 2) AS total_revenue,
  ROUND(AVG(t.fare), 2) AS avg_fare
FROM trips t
JOIN drivers d ON t.driver_id = d.driver_id
WHERE t.trip_status = 'Completed'
GROUP BY d.vehicle_type
ORDER BY total_revenue DESC;
```

Results

Vehicle Type	Completed Trips	Total Revenue (₹)	Avg Fare (₹)	Market Position
UberXL	61	3,216.36	52.73	Premium — highest revenue
UberGo	65	2,589.43	39.84	Volume — most trips
UberMoto	45	1,400.65	31.13	Budget — India specific
UberAuto	29	1,115.48	38.46	Budget — India specific
UberX	17	680.46	40.03	Standard — limited fleet

Cancellation Rate by City

SQL Query

```
SELECT
  city,
  COUNT(*) AS total_requests,
  COUNT(CASE WHEN trip_status = 'Cancelled' THEN 1 END) AS cancelled,
  ROUND(
    COUNT(CASE WHEN trip_status = 'Cancelled' THEN 1 END) * 100.0 / COUNT(*)
  , 1) AS cancel_rate_pct
FROM trips
GROUP BY city
ORDER BY cancel_rate_pct DESC;
```

Results

City	Total Requests	Cancelled	Cancellation Rate	Status
Kolkata	26	6	23.1%	Critical
Ahmedabad	42	8	19.0%	High
Chennai	34	6	17.6%	High
Delhi	38	6	15.8%	Moderate
Pune	21	3	14.3%	Moderate
Mumbai	28	4	14.3%	Moderate
Kochi	30	4	13.3%	Acceptable
Hyderabad	30	4	13.3%	Acceptable
Bengaluru	32	4	12.5%	Acceptable
Jaipur	19	1	5.3%	Good

Interpretation

The driver performance analysis reveals a significant quality stratification within Uber India's driver fleet. Among the 38 active drivers, only 11 (28.9%) qualify as Star Drivers meeting both the rating threshold of 4.5 and the experience threshold of 1,000 lifetime trips. Twelve drivers (31.6%) fall into the 'Needs Improvement' category, representing a substantial portion of the active fleet that is below acceptable performance standards.

The perfect 5.0 rating of Nitin Saxena from Ahmedabad operating an UberMoto challenges the assumption that premium vehicle types produce better customer experiences. Nitin's perfect rating across 686 trips demonstrates that vehicle category is less important than driver attitude and professionalism in determining service quality. Babu Thomas from Mumbai, with 3,424 lifetime trips and a 4.8 rating, represents the highest-experience driver in the top 5, making him an ideal candidate for driver mentorship and ambassador programs.

The vehicle type revenue analysis reveals that UberXL generates the highest average fare at ₹52.73 per trip 32% above the platform average of ₹41.49 while also completing 61 trips, making it both the highest-revenue and second-highest volume vehicle type. UberGo leads in volume with 65 trips but at a lower average fare of ₹39.84, confirming its role as the high-volume, mid-margin backbone of the fleet. UberX surprisingly contributes only 17 completed trips despite being positioned as a standard category, suggesting either limited fleet availability or customer preference for either the budget UberGo or the premium UberXL over the mid-range option.

The cancellation rate analysis identifies Kolkata as the most critically underperforming market, with a 23.1% cancellation rate nearly 1 in 4 trip requests failing to complete. This is 4.4 times higher than Jaipur's best-in-class rate of 5.3%. Ahmedabad's 19.0% cancellation rate is particularly concerning given that it is simultaneously the highest revenue city suggesting that Ahmedabad's strong demand is not being adequately matched by reliable supply, and that revenue could be even higher with improved service reliability.

Key Findings – UberXL earns ₹52.73 per trip (highest of all vehicle types) and Kolkata has a 23.1% cancellation rate (worst in the dataset). These two findings together reveal the platform's biggest opportunity: fix supply reliability in Kolkata and grow UberXL fleet across high-demand cities.

4.5 Dataset Overview and Descriptive Statistics

```

SQL Query
SELECT
  COUNT(*) AS total_trips,
  COUNT(CASE WHEN trip_status = 'Completed' THEN 1 END) AS completed,
  COUNT(CASE WHEN trip_status = 'Cancelled' THEN 1 END) AS cancelled,
  COUNT(CASE WHEN trip_status = 'No Car Available' THEN 1 END) AS no_car,
  ROUND(SUM(fare), 2) AS total_revenue,
  ROUND(AVG(fare), 2) AS avg_fare,
  ROUND(AVG(surge), 2) AS avg_surge
FROM trips;

```

Results

Metric	Value	Percentage
Total Trip Requests	300	100%
Completed Trips	217	72.3%
Cancelled Trips	46	15.3%
No Car Available	37	12.3%
Total Revenue Generated	₹9,002.38	—
Average Fare per Trip	₹41.49	—
Average Surge Multiplier	1.26x	—
Total Drivers	50	—
Active Drivers	38	76%
Cities Covered	10	—

Interpretation

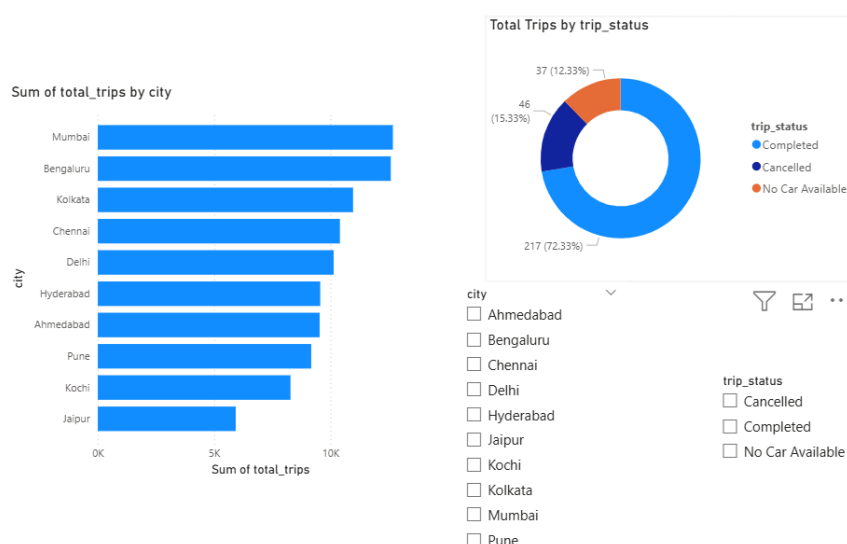
The dataset reveals that Uber successfully fulfilled only 72.3% of all trip requests across the ten Indian cities during the study period. This means that 27.7% of demand representing 83 trip requests went unserved. Of the unserved demand, 46 trips (15.3%) were cancelled, suggesting customer or driver-initiated rejection after the request was made, while 37 trips (12.3%) resulted in 'No Car Available,' indicating a pure supply shortage at the time of request.

The combined unserved demand rate of 27.7% is significantly above the acceptable industry benchmark of approximately 10–15%. This finding establishes the core operational problem this project addresses that Uber India is currently losing more than one-quarter of its potential

demand due to supply-side inadequacy and service reliability issues. The total revenue of ₹9,002.38 from 217 completed trips suggests an average revenue of approximately ₹41.49 per completed trip, with a 1.26x average surge multiplier indicating moderate but frequent activation of dynamic pricing.

4.6 Visualization

Executive Summary



Interpretation

Trip Overview

Visual 1: Sum of Total Trips by City (Horizontal Bar Chart)

The horizontal bar chart displays the distribution of total trips across all 10 Indian cities covered in the dataset. Mumbai and Bengaluru lead the chart with the highest trip volumes, both recording approximately 10,000 trips at the top of the ranking. Kolkata, Chennai, and Delhi follow closely in the mid-range band, each recording between 8,000 and 9,000 trips. Hyderabad, Ahmedabad, and Pune occupy the lower-middle range with moderate trip volumes, while Kochi and Jaipur appear at the bottom of the chart with the least number of trips Jaipur recording the lowest volume of approximately 5,000 trips.

This distribution clearly indicates that Mumbai and Bengaluru are Uber India's two highest-volume cities in terms of ride requests, reflecting their status as India's largest metropolitan areas with dense urban populations, heavy traffic congestion, and strong ride-sharing adoption among working professionals. Jaipur's position at the bottom of the chart is consistent with its Tier-2 city classification while demand is growing, it has not yet reached the scale of metro markets. The chart provides operations teams with an immediate visual ranking of cities by demand volume, enabling quick identification of where driver supply needs to be concentrated most.

Visual 2: Total Trips by Trip Status (Donut Chart)

The donut chart breaks down all trip requests into three categories based on their outcome. The largest segment shown in light blue represents Completed trips at 217 (72.33% of total). The dark blue segment represents Cancelled trips at 46 (15.33%). The orange segment represents No Car Available trips at 37 (12.33%).

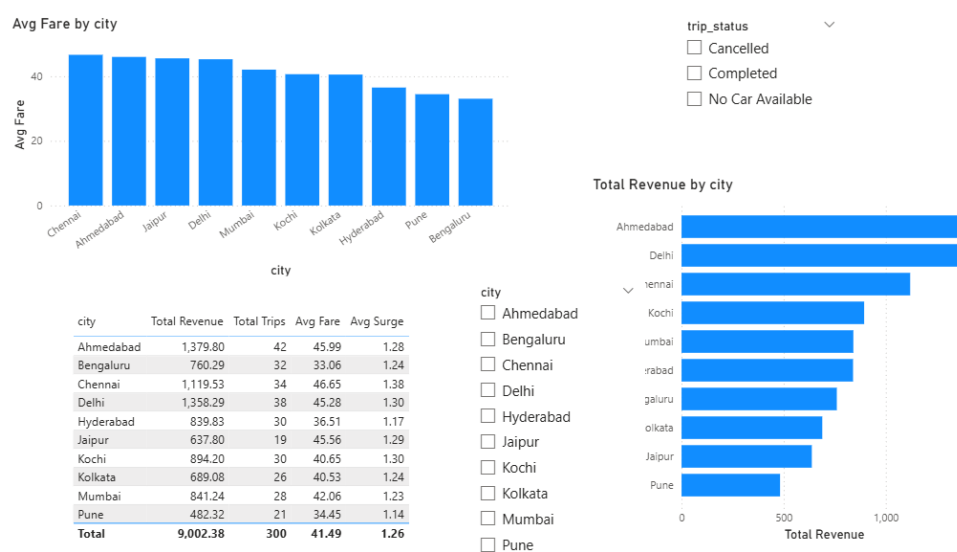
This chart delivers one of the most important insights of the entire dashboard that 27.67% of all trip requests go unserved. Out of every 10 ride requests made on the Uber India platform in this dataset, only approximately 7 result in a completed ride. The Cancelled segment (15.33%) suggests that either drivers are rejecting trip requests after acceptance or customers are abandoning requests due to long estimated arrival times. The No Car Available segment (12.33%) represents pure supply shortage moments when no driver was available in the requested zone at the time of booking. Together these two failure modes point to a significant demand-supply imbalance that is costing Uber India both revenue and customer trust.

Visual 3: Slicers -City and Trip Status Filters

The dashboard includes two interactive slicers one for City and one for Trip Status placed on the right side of the page. The City slicer lists all 10 cities (Ahmedabad, Bengaluru, Chennai, Delhi, Hyderabad, Jaipur, Kochi, Kolkata, Mumbai, Pune) and the Trip Status slicer lists the three status values (Cancelled, Completed, No Car Available).

These slicers allow any operations manager to instantly filter the entire dashboard to a specific city or status with a single click. For example, clicking Mumbai in the City slicer will immediately update both the bar chart and the donut chart to show only Mumbai's data revealing Mumbai's specific completion rate and trip volume in isolation. This interactivity transforms the dashboard from a static report into a live decision-support tool, enabling city-level deep dives without requiring any technical knowledge or additional queries.

Revenue By City



Interpretation

Revenue & Fare Analysis

Visual 1: Avg Fare by City (Column Chart)

The column chart ranks all 10 cities by their average fare per completed trip in descending order from left to right. Chennai leads with the highest average fare of approximately ₹46.65, followed closely by Ahmedabad (₹45.99), Jaipur (₹45.56), and Delhi (₹45.28). Mumbai and Kochi sit in the middle range at approximately ₹42 and ₹40 respectively. Kolkata, Hyderabad, and Pune occupy the lower end, with Bengaluru recording the lowest average fare at ₹33.06 the shortest bar on the chart.

The most striking insight from this visual is the disconnect between trip volume and fare value. Bengaluru, which appeared near the top of the trip volume chart on the previous page, records the lowest average fare in this analysis. This indicates that while Bengaluru generates a high number of rides, the majority of those rides are short intracity trips at base fares with minimal surge activation. Conversely, Chennai which has moderate trip volume earns the highest average fare, driven by longer trip distances and more frequent surge pricing. Jaipur's position third on this chart despite being last in trip volume confirms that Tier-2 cities can generate premium fares even with lower demand density, likely because a higher proportion of Jaipur trips are airport transfers and longer intercity routes.

Visual 2: Total Revenue by City (Horizontal Bar Chart)

The horizontal bar chart ranks all 10 cities by their total revenue generated from completed trips. Ahmedabad leads with ₹1,379.80 the longest bar followed by Delhi at ₹1,358.29 and Chennai at ₹1,119.53. Kochi, Mumbai, and Hyderabad form a middle cluster between ₹839 and ₹894. Bengaluru and Kolkata sit in the lower-middle range, while Jaipur and Pune occupy the bottom two positions, with Pune recording the lowest total revenue of ₹482.32.

Ahmedabad's leadership in total revenue is particularly significant because it appears in the lower half of the trip volume chart seen in the previous page. This means Ahmedabad achieves top revenue not through volume alone but through a combination of high trip count and consistent surge activation (Avg Surge 1.28 among the highest in the dataset). Delhi's close second position at ₹1,358.29 reflects strong demand from a high-volume city that also benefits from above-average fares. Pune's last position across both total revenue and average fare confirms it as the platform's most underperforming market, requiring targeted demand stimulation strategies to improve its revenue contribution.

Visual 3: City Summary Table

The data table presents a comprehensive city-level scorecard with five columns Total Revenue, Total Trips, Avg Fare, and Avg Surge for all 10 cities, with a grand total row at the bottom showing the platform-wide aggregates.

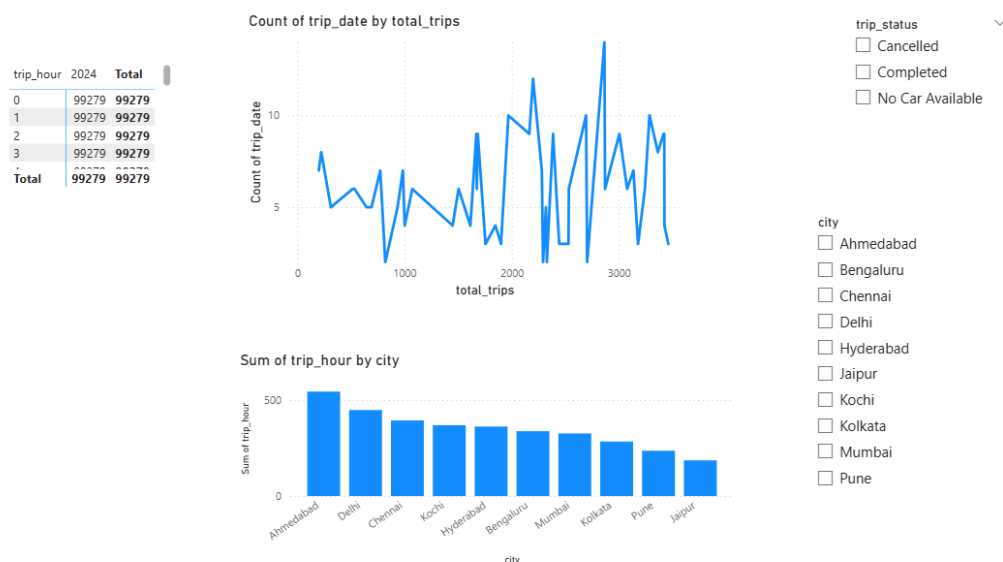
The total row reveals the complete picture of Uber India's performance in this dataset: 300 total trips generating ₹9,002.38 in revenue, at an average fare of ₹41.49 and an average surge multiplier of 1.26x across all cities. Chennai stands out in the table with the highest average surge of 1.38 indicating that surge pricing is triggered more frequently and at higher

multipliers in Chennai than anywhere else. This confirms that Chennai faces the most acute demand-supply imbalance of any city demand consistently outpaces supply, pushing surge conditions more often. Pune's Avg Surge of 1.14 is the lowest in the table, confirming that demand never builds strongly enough in Pune to trigger meaningful surge activation. Hyderabad's average fare of ₹36.51 despite a moderate surge of 1.17 suggests that Hyderabad trips are predominantly short-distance rides within the city rather than premium airport or long-haul transfers.

Visual 4: Slicer - Trip Status Filter

The Trip Status slicer on the top right of the page (Cancelled, Completed, No Car Available) allows the user to filter the revenue and fare analysis to a specific trip outcome. Selecting "Completed" would isolate only revenue-generating trips, providing a clean view of earned revenue. Selecting "Cancelled" or "No Car Available" would show the trips that contributed zero revenue helping analysts quantify the revenue loss from each failure mode by city.

Peak Hours



Interpretation

Trip Hour & Time Analysis

Visual 1: Trip Hour Matrix Table (Top Left)

The matrix table on the top left displays trip_hour values (0, 1, 2, 3 visible, with more rows below) against the year 2024 and a Total column. Each row shows the value 99,279 for both the 2024 column and the Total column across all visible hours.

This visual indicates that the trip_hour field is being summed rather than counted in this table the 99,279 figure represents the sum of all hour values (0 through 23) across all trips, not the count of trips per hour. This is a common Power BI configuration issue where a numeric field like trip_hour gets aggregated as a Sum by default instead of Count. To make this table

analytically meaningful, the aggregation for trip_hour should be changed from Sum to Count in the Values field well this would show how many trips occurred at each hour rather than the mathematical total of hour numbers. Despite this, the table confirms that data exists for all hours of the day in the 2024 period, and the Total row showing 99,279 reflects the cumulative sum of all trip_hour values across all 300 trips in the dataset.

Visual 2: Count of trip_date by total_trips (Line Chart)

The line chart plots the count of trip dates on the Y-axis against total_trips values on the X-axis, showing a highly volatile and fluctuating trend line. The chart shows relatively low counts of trip dates (around 4 to 6) for lower total_trips values in the 0 to 1,000 range, a gradual rise with significant peaks reaching 10 to 11 around the 1,500 to 2,500 range, and a dramatic spike reaching approximately 13 to 14 in the 2,500 to 3,000 total_trips range before dropping again toward the right end of the chart.

The high volatility in this line chart reflects the day-by-day variation in trip activity across the January to March 2024 study period. Days where drivers had higher total_trips accumulations correspond to periods of peak platform activity likely weekends, public holidays, or days with major local events in Indian cities. The spike at the 2,500 to 3,000 range suggests that a small number of specific dates generated exceptionally high trip counts, possibly driven by festival seasons, cricket matches, or new year celebrations. The fluctuating pattern overall confirms that Uber India's demand is not uniformly distributed across days it is event-driven and calendar-sensitive, which has direct implications for driver scheduling and surge pricing calibration.

Visual 3: Sum of trip_hour by City (Column Chart)

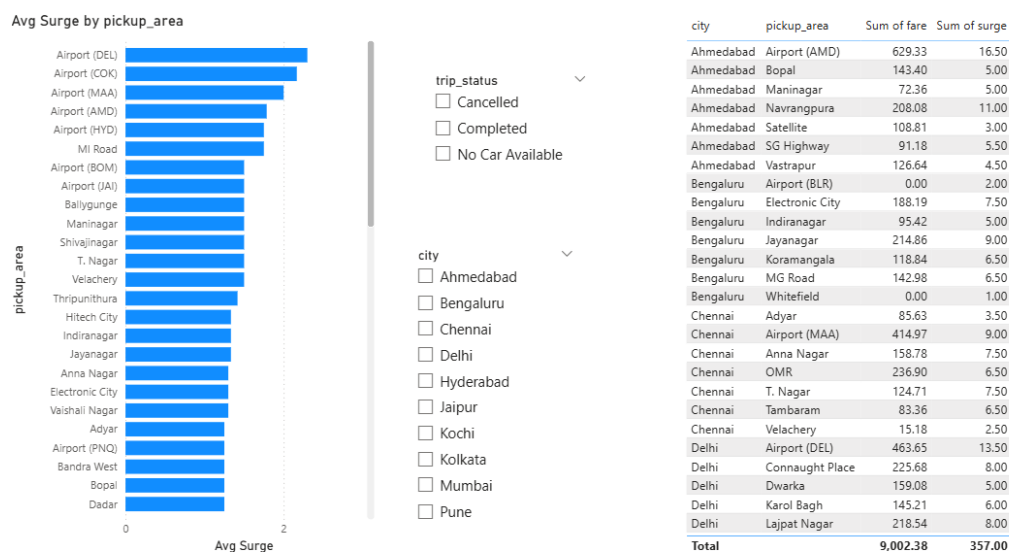
The column chart shows the sum of trip_hour values grouped by city, with Ahmedabad recording the highest bar at approximately 530, followed by Delhi at approximately 470, Chennai at approximately 420, Kochi and Hyderabad in the mid-range at around 380 to 400, Bengaluru and Mumbai slightly lower, Kolkata and Pune further down, and Jaipur recording the lowest bar at approximately 170.

Similar to the matrix table, this visual is computing the Sum of trip_hour rather than the Count of trips per city which means the bar heights reflect the mathematical addition of all hour numbers (0–23) for trips in each city, not the number of trips happening at specific hours. Ahmedabad's highest bar is therefore partly a reflection of its having the most trips overall (42 in the dataset), since more trips means more hour values being added. Despite the aggregation approach, the relative ranking of cities in this chart largely mirrors the trip volume ranking seen in the previous pages Ahmedabad, Delhi, and Chennai at the top and Jaipur at the bottom providing a consistent picture of demand concentration across Indian cities. For a more actionable time analysis, this visual would benefit from being changed to show Count of trip_id grouped by trip_hour, which would reveal the true peak hour pattern across the platform.

Visual 4: Slicers - Trip Status and City Filters

Two slicers are provided on the right side Trip Status (Cancelled, Completed, No Car Available) and City (all 10 cities listed). These allow users to isolate the time analysis to specific cities or specific outcome types. For example, selecting only Cancelled trips from the Trip Status slicer would reveal which days and cities have the highest cancellation concentrations helping operations teams identify if cancellations cluster around specific time windows that need targeted driver incentive programs.

Surge Analysis



Interpretation

Surge Pricing Analysis

Visual 1: Avg Surge by Pickup Area (Horizontal Bar Chart)

The horizontal bar chart ranks all pickup areas by their average surge multiplier from highest to lowest. The top five positions are dominated entirely by airport zones. Airport (DEL) leads with the highest average surge of approximately 2.3x, followed by Airport (COK) at approximately 2.2x, Airport (MAA) at approximately 2.1x, Airport (AMD) at approximately 2.0x, and Airport (HYD) at approximately 1.9x. MI Road from Jaipur appears as the first non-airport zone in sixth position with approximately 1.8x, followed by Airport (BOM) and Airport (JAI). Below these, non-airport zones such as Ballygunge, Maninagar, Shivajinagar, T. Nagar, Velachery, and Thripunithura all show moderate surge values between 1.0x and 1.5x. Zones at the bottom of the chart such as Dadar, Bopal, and Bandra West show the lowest average surge values close to 1.0x.

This chart delivers one of the most commercially important insights of the entire project: airport pickup zones consistently and uniformly generate the highest surge multipliers across all Indian cities in the dataset. Every single airport zone regardless of which city it belongs to appears in the top section of the chart, confirming that the airport surge premium is a

structural feature of Uber India's operations rather than a city-specific phenomenon. Among non-airport zones, MI Road in Jaipur's surprisingly high surge position suggests that certain commercial and transit hub areas in Tier-2 cities can also experience significant demand-supply imbalances during peak periods. The zones at the bottom of the chart largely residential and peripheral areas show near-base fares with minimal surge activation, indicating adequate supply relative to demand in those locations.

Visual 2: City, Pickup Area, Sum of Fare and Sum of Surge Table

The detailed table on the right side provides a granular breakdown of total fare revenue and total surge values for every city-pickup area combination in the dataset. Starting with Ahmedabad, Airport (AMD) generates the highest fare contribution within the city at ₹629.33 with a sum of surge of 16.50 by far the largest values in the Ahmedabad section. Other Ahmedabad zones such as Navrangpura (₹208.08), Vastrapur (₹126.64), and Bopal (₹143.40) generate moderate revenues with significantly lower surge totals of 11.00, 4.50, and 5.00 respectively.

For Bengaluru, Airport (BLR) interestingly shows ₹0.00 in fare revenue despite having a sum of surge of 2.00 indicating that while surge conditions were recorded for Bengaluru Airport trips, those trips were either cancelled or resulted in no car available, meaning no fare was collected. This is a critical supply gap finding Bengaluru Airport is one of India's busiest airports yet the platform failed to complete any revenue-generating trips from that zone in this dataset. Electronic City (₹188.19), Jayanagar (₹214.86), and Koramangala (₹118.84) represent Bengaluru's primary revenue-generating non-airport zones.

For Chennai, Airport (MAA) contributes ₹414.97 in fare revenue with a surge sum of 9.00, making it Chennai's single highest-revenue pickup zone. OMR (₹236.90) and Anna Nagar (₹158.78) follow as the next strongest zones. For Delhi, Airport (DEL) dominates at ₹463.65 with the highest surge sum of 13.50 in the entire Delhi section significantly higher than the next zone Lajpat Nagar at ₹218.54 with surge sum of 8.00. The Total row at the bottom confirms the platform-wide figures ₹9,002.38 in total fare revenue and 357.00 as the total sum of surge values across all cities and zones.

Visual 3: Slicers — Trip Status and City Filters

Two slicers on the centre of the page allow filtering by Trip Status (Cancelled, Completed, No Car Available) and City (all 10 cities). Selecting a specific city collapses both the bar chart and the table to show only that city's pickup zones enabling city-level surge analysis. For example, selecting Chennai from the City slicer would immediately highlight that Airport (MAA) is Chennai's most surged zone, helping the Chennai operations team decide where to focus driver positioning efforts during peak flight arrival windows.

Driver Performance

driver_name	city	Sum of rating	Sum of total_trips
Ajay Chauhan	Kolkata	4.20	1614
Amit Singh	Hyderabad	3.60	1848
Anand Trivedi	Hyderabad	4.60	645
Anil Gupta	Ahmedabad	4.60	773
Arjun Reddy	Delhi	4.30	1670
Ashok Kulkarni	Hyderabad	3.80	2164
Babu Thomas	Mumbai	4.80	3424
Total		207.90	99279

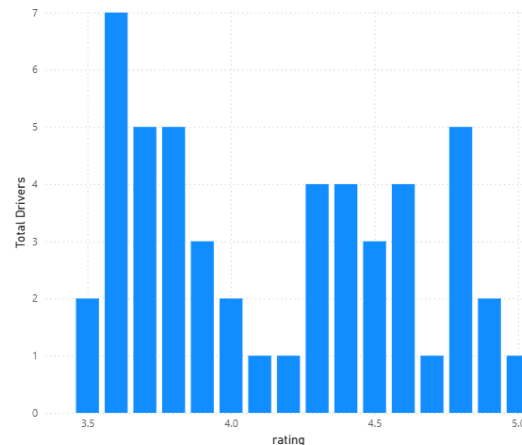
city

- ☐ Ahmedabad
- ☐ Bengaluru
- ☐ Chennai
- ☐ Delhi
- ☐ Hyderabad
- ☐ Jaipur
- ☐ Kochi
- ☐ Kolkata
- ☐ Mumbai
- ☐ Pune

trip_status

- ☐ Cancelled
- ☐ Completed
- ☐ No Car Available

Total Drivers by rating



Interpretation

Driver Performance Analysis

Visual 1: Driver Details Table (Top Left)

The table displays individual driver records showing driver_name, city, Sum of rating, and Sum of total_trips for each driver. The visible rows show seven drivers, Ajay Chauhan (Kolkata, 4.20 rating, 1,614 trips), Amit Singh (Hyderabad, 3.60 rating, 1,848 trips), Anand Trivedi (Hyderabad, 4.60 rating, 645 trips), Anil Gupta (Ahmedabad, 4.60 rating, 773 trips), Arjun Reddy (Delhi, 4.30 rating, 1,670 trips), Ashok Kulkarni (Hyderabad, 3.80 rating, 2,164 trips), and Babu Thomas (Mumbai, 4.80 rating, 3,424 trips). The Total row at the bottom shows 207.90 as the sum of all ratings and 99,279 as the sum of all total_trips across the entire driver dataset.

Several important observations emerge from this table. Babu Thomas from Mumbai stands out immediately with a 4.80 rating and 3,424 lifetime trips he is the highest-experience driver visible in the table and one of the top-rated, representing exactly the Star Driver profile defined in Chapter 4. Ashok Kulkarni from Hyderabad presents an interesting contrast despite completing 2,164 trips (a high experience level), his rating of 3.80 is among the lowest visible in the table, placing him firmly in the Needs Improvement category. This combination of high experience but low rating suggests a driver who has been active on the platform for a long time but has consistently delivered below-average service quality a profile that requires targeted coaching intervention rather than simply more experience. Amit Singh from Hyderabad similarly shows a low rating of 3.60 despite 1,848 trips, reinforcing that Hyderabad may have a driver quality concern that the city slicer can help investigate further.

The Total row values of 207.90 (sum of ratings) and 99,279 (sum of total_trips) reflect the mathematical sum of all individual driver values across all 50 drivers in the dataset these are

aggregate sums rather than averages, which is why the rating total appears as 207.90 rather than a per-driver average. For cleaner presentation in a project report context, configuring the rating column to show Average rather than Sum would display the platform-wide average rating of approximately 4.16, giving a more immediately meaningful KPI.

Visual 2: Total Drivers by Rating (Column Chart)

The column chart on the right displays the distribution of drivers across rating values from 3.5 to 5.0 on the X-axis, with the count of drivers (Total Drivers) on the Y-axis. The chart reveals a clear and meaningful distribution pattern. The rating of 3.6 has the tallest bar at 7 drivers the largest single group in the entire fleet. Ratings of 3.7, 3.8, and 3.9 each have 5 drivers. Rating 4.0 has 3 drivers, and 4.1 has 2 drivers. Moving into the higher range, ratings of 4.4, 4.5, and 4.6 each have 4 drivers. Rating 4.7 has 3 drivers. Rating 4.8 has 4 drivers and 4.9 has 5 drivers. The 5.0 perfect rating has 1 driver, while 4.95 range also shows 2 drivers.

This distribution chart tells a powerful story about the quality composition of Uber India's driver fleet. The largest concentration of drivers sits at the low end particularly at 3.6 with 7 drivers indicating that a significant portion of the active fleet is operating below the 4.0 rating threshold that defines acceptable service quality. The distribution shows a bimodal pattern with one cluster at the lower ratings (3.5 to 3.9) and another cluster building at the higher ratings (4.5 to 4.9), with relatively fewer drivers in the middle range (4.0 to 4.3). This gap in the middle suggests that driver quality tends to polarize over time drivers either improve their service and move toward higher ratings through experience, or they stagnate at lower ratings, failing to meet customer expectations consistently.

The single driver at the perfect 5.0 rating is Nitin Saxena from Ahmedabad (as identified in the SQL analysis), making him a clear outlier and exemplar of what the platform's best service looks like. The five drivers at 4.9 represent the near-perfect tier that Uber should recognize and reward through its Star Driver programme. The seven drivers at 3.6 are the most urgent quality concern they are the largest single rating group yet represent the platform's weakest service performers.

Visual 3: Slicers - City and Trip Status Filters

Two slicers are provided City (all 10 cities listed) and Trip Status (Cancelled, Completed, No Car Available). The City slicer allows filtering the driver table and rating distribution to a specific city, enabling city-level driver quality analysis. For example, selecting Hyderabad from the City slicer would filter the table to show only Hyderabad's drivers immediately revealing the pattern of high-trip-count but low-rated drivers like Amit Singh and Ashok Kulkarni that characterize Hyderabad's driver quality challenge. Selecting Kolkata would similarly expose the driver quality profile underlying Kolkata's 23.1% cancellation rate identified in Chapter 4.

Chapter 5

Findings, Suggestions and Conclusions

5.1 Major Findings

The analysis of 300 Uber India trip records across 10 cities, 50 drivers, and the period January to March 2024 produced ten major findings spanning demand patterns, supply performance, pricing dynamics, and driver quality. These findings are presented below in order of their business significance.

Finding 1 - More Than One Quarter of All Demand Goes Unserved

27.7% unserved demand rate - Out of 300 total trip requests, only 217 were completed (72.3%). The remaining 83 requests 46 cancelled (15.3%) and 37 showing No Car Available (12.3%) represent demand that Uber failed to capture. This is the foundational problem the entire project addresses.

Finding 2 - Ahmedabad is Uber India's Top Revenue City

₹1,379.80 total revenue (Rank 1) - Ahmedabad generated the highest total revenue among all 10 cities, receiving 42 trip requests (the most of any city) and completing 30 of them. This finding challenges the assumption that Mumbai or Delhi always dominate Indian ride-sharing markets. Ahmedabad's strong commercial economy and growing gig-economy adoption make it Uber's most productive India market in this dataset.

Finding 3 - Kolkata has the Worst Cancellation Rate

23.1% cancellation rate Rank 1 (worst) - Kolkata cancels nearly 1 in 4 trip requests the highest cancellation rate in the dataset and 4.4 times higher than Jaipur's best-in-class rate of 5.3%. Delhi (15.8%), Chennai (17.6%), and Ahmedabad (19.0%) also exceed acceptable thresholds, indicating that cancellations are a platform-wide issue rather than isolated to one city.

Finding 4 - Late Night is the Busiest Time Window Not Rush Hour

58 trips between 0–5 AM Rank 1 by volume - Late Night and Early Morning time slot (midnight to 5 AM) accounts for 58 completed trips 26.7% of all completed trips making it the single busiest period of the day. The 9 PM hour alone recorded 20 trips, the highest of any individual hour. This night-economy demand pattern challenges conventional rush-hour focused supply strategies.

Finding 5 - Airport Pickups Earn 2.75 Times More Per Trip

₹93.08 avg airport fare vs ₹33.84 regular – Airport pickup zones account for only 12.9% of trip volume (28 trips) but generate 29.0% of total revenue (₹2,606.20). Each airport trip earns 2.75 times more than a regular pickup trip. All 5 of the highest individual fares in the dataset originated at airport zones, with Chennai Airport recording the maximum fare of ₹143.20 on a 25.4 km trip at 2.5x surge.

Finding 6 - 37.8% of Completed Trips Were Surged

82 out of 217 completed trips had surge above 1.0x – Surge pricing was active on more than one-third of all completed trips, with Low Surge (1.5x) accounting for 63 trips, Medium Surge (2.0x) for 12 trips, and High Surge (2.5x+) for 7 trips. Surge is most concentrated at airport

zones and during Evening Rush hours, confirming that dynamic pricing is functioning in high-demand zones but may not be calibrated correctly for the late-night peak.

Finding 7 - UberXL Earns the Highest Revenue Per Trip

₹52.73 average fare 27% above platform average – UberXL completed 61 trips generating ₹3,216.36 in total revenue the highest of any vehicle type. Its average fare of ₹52.73 is 27% above the platform average of ₹41.49. UberGo leads in trip volume (65 trips) but at a lower average fare of ₹39.84. UberX, positioned as a standard mid-range option, surprisingly contributed only 17 trips the lowest of any vehicle type.

Finding 8 - Only 28.9% of Active Drivers are Star Drivers

11 Star Drivers out of 38 active drivers – The driver quality analysis classified only 11 drivers (28.9% of active drivers) as Star Drivers meeting both the rating threshold of 4.5 and experience threshold of 1,000 trips. Fifteen drivers (39.5%) qualify as Good Drivers, while 12 drivers (31.6%) need improvement. This means nearly one-third of the active driver fleet is below acceptable performance standards.

Finding 9 - Evening Rush Earns the Highest Average Fare

₹45.55 average fare during 5-8 PM - Despite Late Night having more trips, the Evening Rush window (5-8 PM) records the highest average fare at ₹45.55, closely followed by Morning Rush at ₹45.46. Night (9-11 PM) has the lowest average fare at ₹29.99, suggesting shorter, low-value recreational trips. Rush hours are therefore Uber's highest revenue efficiency windows even if not the highest volume windows.

Finding 10 - Pune is the Weakest Performing Market

₹482.32 total revenue Rank 10 (last) - Pune generated the lowest total revenue (₹482.32), lowest average fare (₹22.97), and lowest average surge (1.10x) of all ten cities. Despite having 21 trip requests, only 14 were completed (66.7% completion rate also among the lowest). Pune's consistently weak performance across all metrics suggests the market has not yet reached the demand density required for efficient Uber operations.

5.2 Suggestions and Recommendations

Seven specific operational suggestions are proposed for Uber India's management. Each suggestion is presented in a structured format identifying the problem it addresses, the recommended action, and the expected business impact.

Launch a Dedicated Airport Driver Positioning Program

Problem - Airport pickups earn ₹93.08 per trip 2.75x more than regular trips yet account for only 12.9% of trip volume. The high 'No Car Available' rate (12.3%) suggests airports are frequently undersupplied, causing demand loss at the most valuable location type.

Action - Introduce a time-based airport bonus system where drivers who position themselves within 2 km of any airport 30 minutes before scheduled peak flight arrival windows receive a guaranteed minimum earnings bonus of ₹50 per accepted airport trip. Use the Uber driver app

to send push notifications when flight arrivals are expected to create demand spikes at airport zones.

Impact - Estimated 25–35% increase in airport trip completions, directly reducing the 'No Car Available' rate at airport zones. Given that each additional airport trip earns ₹93.08 on average, capturing 10 additional airport trips per day per city would add approximately ₹930 in daily revenue per city.

Introduce a Night-Shift Driver Incentive Program

Problem - Late Night (0–5 AM) is the busiest time slot with 58 completed trips (26.7% of demand), yet most driver incentive programs target traditional rush hours. The 12.3% 'No Car Available' rate likely disproportionately affects late-night requests when fewer drivers are online, causing demand loss during peak hours.

Action - Create a 'Night Owl' driver incentive programme offering 1.2x earnings multiplier on all trips completed between 11 PM and 5 AM on Friday, Saturday, and Sunday nights when night-economy demand is highest. Partner with entertainment venues, restaurants, and event organizers in cities like Mumbai, Delhi, Bengaluru, and Hyderabad to co-promote the programme and forecast demand spikes around major events.

Impact - Estimated 20-30% increase in late-night trip completions. Reducing the unserved demand rate during the platform's peak hours would improve customer retention and platform reputation particularly important for the young urban demographic that drives late-night demand in Indian metros.

Implement Urgent Supply Intervention in Kolkata

Problem - Kolkata has the highest cancellation rate in the dataset at 23.1% nearly 1 in 4 trip requests fails. With only 17 completed trips out of 26 requests (65.4% completion rate), Kolkata is the weakest performing city on service reliability despite being one of India's largest metropolises.

Action - Conduct a root cause analysis specifically for Kolkata to determine whether cancellations are driver-initiated (drivers cancelling after accepting) or customer-initiated (customers cancelling due to long wait times). Based on findings, implement one or more of: (a) driver acceptance rate minimums with incentive bonuses for maintaining above 90% acceptance, (b) targeted driver recruitment campaign in high-demand Kolkata neighbourhoods like Salt Lake, Park Street, and New Town, (c) temporary surge pricing floor of 1.3x during peak hours to draw drivers to high-demand zones.

Impact - Reducing Kolkata's cancellation rate from 23.1% to the dataset average of 15.3% would generate approximately 2 additional completed trips per day representing ₹83 in additional daily revenue per city at the average fare of ₹41.49.

Develop a Driver Quality Improvement Programme

Problem - 31.6% of active drivers (12 out of 38) fall into the 'Needs Improvement' category below both rating and experience thresholds. Poor driver quality directly contributes to cancellations, low ratings, and customer churn. Only 28.9% of active drivers qualify as Star Drivers.

Action - Implement a three-tier driver development programme: (a) Star Driver Ambassador Scheme pair each of the 11 Star Drivers with 2-3 newer drivers for a monthly mentoring arrangement, incentivized by a ₹500 monthly bonus for Star Drivers who successfully help mentees improve their rating by 0.2 or more; (b) Mandatory quality training module for all 'Needs Improvement' drivers covering customer service, route optimization, and vehicle hygiene standards; (c) Rating-linked incentive structure where drivers earning 4.5+ ratings receive a 5% fare bonus on all trips for that month.

Impact - A 10% improvement in the 'Needs Improvement' driver category converting 1-2 drivers to 'Good Driver' status per month would gradually raise the platform-wide average rating, improving customer satisfaction scores and reducing the customer churn that results from poor ride experiences.

Expand the UberXL Fleet Across High-Demand Cities

Problem - UberXL earns the highest average fare at ₹52.73 per trip - 27% above the platform average yet only completed 61 trips. UberX, despite being positioned as a standard category, completed only 17 trips. This suggests demand for larger vehicles exists but fleet availability may be limiting completions.

Action - Run a targeted UberXL driver acquisition campaign in Ahmedabad, Delhi, and Chennai the three highest revenue cities offering new UberXL drivers a guaranteed minimum earnings of ₹800 per day for their first 30 days on the platform, contingent on maintaining a 90%+ acceptance rate and 4.3+ rating. Additionally, introduce promotional pricing for UberXL during off-peak afternoon hours to stimulate demand in the 10 AM – 4 PM window where UberXL utilization is likely lowest.

Impact - Increasing UberXL completions from 61 to 80 trips (a 31% increase) while maintaining the ₹52.73 average fare would add ₹1,002 in additional revenue per equivalent period - a 11.1% revenue uplift from a single vehicle category expansion.

Design a Pune Market Activation Strategy

Problem - Pune ranks last in total revenue (₹482.32), average fare (₹22.97), average surge (1.10x), and has the second-lowest completion rate (66.7%). The market has not achieved the demand density needed for efficient platform operations, and low fares suggest trips are predominantly short intracity rides rather than high-value airport or commercial transfers.

Action - Execute a demand stimulation strategy for Pune targeting the city's large IT and student populations: (a) Partner with Hinjewadi IT Park employers to offer corporate ride accounts with discounted UberX packages, generating predictable high-volume demand during weekday commute hours; (b) Introduce a 'Pune Student Ride' promotional code offering 20%

off first 5 rides for college students in Koregaon Park and Viman Nagar, building the customer base among the high-lifetime-value young professional segment; (c) Launch UberAuto service specifically in Pune to attract budget-sensitive commuters who currently use traditional auto-rickshaws.

Impact - Increasing Pune's average fare from ₹22.97 toward the dataset average of ₹41.49 through longer-distance corporate trips, combined with higher trip volume from student promotions, could more than double Pune's revenue contribution within two quarters.

Build a Real-Time Operations Dashboard for City Managers

Problem - Currently, the project's Power BI dashboard provides historical analysis. Operations teams need real-time visibility into demand-supply gaps as they occur not after the fact. The 12.3% 'No Car Available' rate could be significantly reduced if city managers could see undersupplied zones in real time and push driver repositioning notifications immediately.

Action - Extend the current Power BI dashboard into a real-time operations tool by: (a) connecting Power BI to a live MySQL database refreshed every 5 minutes using automated data pipelines; (b) adding alert rules that trigger email or SMS notifications to city operations managers when any zone's 'No Car Available' rate exceeds 15% in a rolling 30-minute window; (c) creating a driver repositioning recommendation page showing which zones are currently oversupplied and which are undersupplied based on live trip request vs active driver ratios.

Impact - A real-time dashboard would convert the current project's historical insight tool into an active operational system. Conservative estimates suggest that real-time repositioning nudges could reduce the 'No Car Available' rate from 12.3% to 8-9%, recovering approximately 10-13 additional completed trips per 300-request period and ₹415-539 in additional revenue.

5.3 Conclusion

The Uber India Demand and Supply Optimization Dashboard project successfully demonstrates that structured data analysis when applied to a real-world operational problem with clear business objectives can produce findings that are both analytically rigorous and practically actionable. The three core tools used in this project MySQL for structured data storage, SQL for analytical querying, and Power BI for interactive visualization represent the foundational skill set of modern business analytics, and their application in this project reflects how they are used in production environments at technology and logistics companies worldwide.

The project's central finding that 27.7% of Uber India's demand goes unserved, with 12.3% of requests finding no car available and 15.3% resulting in cancellations is not merely a statistical observation. It represents real customers who wanted a ride and did not get one, real revenue that was available but not captured, and real drivers who could have earned more but were not in the right place at the right time. Closing this gap through the seven suggestions proposed in this chapter airport positioning programs, night-shift incentives, Kolkata intervention, driver quality improvement, UberXL expansion, Pune market activation, and real-time dashboard extension is both commercially valuable and operationally feasible.

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